GHAZI UNIVERSITY
DERA GHAZI KHAN

Department: BS Computer Science

Subject: Programming Fundamentals Lab

Lab No: 4

Student Name: Mohammad Sharjeel

Roll No: 12

Section: C

Assignment Submission — Fall 2024

Date ___/___/___

LAB NO 4

Task No 1 write a program to print the number entered by the user. If the number is positive, print positive.

Input

```
#include <Iostream>
Using namespace std;
int main()
{
int num;
cout<<" Enter a number: ";
      cin >> num;
if (num>0)
{
cout <<" The number is positive.";
   }
else if(num<0)
  cout<< The number is negative.";
  }
  else
{
cout<< " The number is zero.";
}
  return 0;
}
```

Output

```
Enter a number: 1
The number is positive.
```

HERO NOTES

TASK NO 2 write a program to check wheter an integer is positive, negative or zero.

Input

```
int main()
{
 int num;
cout << " Enter an integer:";
      cin >> num;
  if (num>0)
{
 cout << " Positive number";
else if (num<0)
 {
cout<< " Negative number";
   }
    else
{
  cout<< " The number is zero";
 }
 return 0;
}
```

OUTPUT

OUT PUT 8= positive

TASK NO 03

calculate student grade:

Marks ≥75 grade A

Marks ≥60 and <75: Grade B

Marks ≥ 45 and <60: GradeC

Marks < 45 : gradeD

Input

```
int main()
{
int marks;
cout<< "Enter your marks:";
    cin >> marks;
if (marks >= 75) {
  cout<<" grade A"; }
else if (marks >= 60) {
    cout<<" Grade B';
}
else if (marks >= 45) {
  cout<<' Grade C';
}
else {
 cout<<" Grade D";
}
return 0;
}
```

Output

Enter your marks 80

Grade A

HERO NOTES

Task 04

Input

```
# include < Iostream>
 Using name std;
 int main()
int a, b, c, num, count = 0;
 Cout<< "Enter three numbers:";
  Cin>>a >>b>>c;
  if(a>b){
if (a>c){
cout<< largest number is "<<a <<endl;
 }
 else{
cout<< "largest number is"<<<<<<endl;
 }
}
else{
number is "<<c <<end1;
  }
}
cout<< " Enter a number to count digits :";
   Cin >> num;
if (num>0){
while (num>0){
  cout++;
num = num/10; }
```

OUTPUT

```
Enter three number:
34, 45, 56
largest number: 56
Enter a number
to count digits:
1234
Number OF
digits: 4
```

HERO NOTES

Coding C++

Auto saved at 19:25:30

RUN MENU

```cpp

#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter a number: ";
    cin >> num;

    if (num > 0)
    {
        cout << "The number is positive.";
    }
    else if (num < 0)
    {
        cout << "The number is negative.";
    }
    else
    {
        cout << "The number is zero.";
    }

    return 0;
}

```

Compile Result

```
Enter a number: 1
The number is positive.
[Process completed - press Enter]
```

Coding C++

Auto saved at 19:42:34

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;

    cout << "Enter an integer: ";
    cin >> num;

    if (num > 0) {
        cout << "Number is positive";
    }
    else if (num < 0) {
        cout << "Number is negative";
    }
    else {
        cout << "Number is zero";
    }

    return 0;
}
```

Compile Result

```
Enter an integer: 8
Number is positive
[Process completed - press Enter]
```

Coding C++

Auto saved at 19:52:32

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main() {
    int marks;

    cout << "Enter your marks: ";
    cin >> marks;

    if (marks >= 75) {
        cout << "Grade A";
    }
    else if (marks >= 60) {
        cout << "Grade B";
    }
    else if (marks >= 45) {
        cout << "Grade C";
    }
    else {
        cout << "Grade D";
    }

    return 0;
}
```



Compile Result

```
Enter your marks: 80
Grade A
[Process completed - press Enter]
```

Coding C++

Nothing changed

RUN MENU

```

#include <iostream>
using namespace std;

int main() {
    int a, b, c, num, count = 0;

    cout << "Enter three numbers: ";
    cin >> a >> b >> c;

    if (a > b) {
        if (a > c) {
            cout << "Largest number is: " << a << endl;
        }
        else {
            cout << "Largest number is: " << c << endl;
        }
    }
    else {
        if (b > c) {
            cout << "Largest number is: " << b << endl;
        }
        else {
            cout << "Largest number is: " << c << endl;
        }
    }

    cout << "Enter a number to count digits: ";
    cin >> num;

    if (num > 0) {
        while (num > 0) {
            count++;
            num = num / 10;
        }

        cout << "Number of digits: " << count;
    }
    else {
        cout << "Number is not positive";
    }

    return 0;
}
```



Compile Result

```
Enter three numbers: 34 45 56
Largest number is: 56
Enter a number to count digits: 1234
Number of digits: 4
[Process completed - press Enter]
```